What makes motion hard to stop?

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Compare momentum changes in a controlled stopping situation.

SUCCESS CRITERIA - I can...

1. Compare mass and velocity while controlling other variables.
2. Use momentum change to compare stopping force for equal times.
3. Avoid claiming that momentum alone determines damage.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

Two ideal Sun-orbits have a_B=**3** a_A.
1. Find T_B/T_A.
2. **Explain why the same normalized** Sun-orbit equation is not automatically valid around another body.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $p = mv$; $F_{avg} = \Delta p/\Delta t$.
Momentum p includes direction. For a fixed stopping time, greater momentum change needs greater average net force.

Worked reasoning: A **2 kg** cart and a **4 kg** cart move right at **3 m/s**. Their momenta mv are **6** and **12 kg·m/s**. Stopping the second requires twice the momentum change.

Your turn: 1. **Draw the velocity** and momentum arrows for both carts.
2. **Explain which pair has equal** lengths.

Before	After

Label mass, velocity and momentum.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Carts A and B both move right at **2 m/s**. A has mass **1 kg**; B has mass **3 kg**. Which has more momentum? **State the controlled quantity**.



A: 1 kg; 2 m/s B: 3 kg; 2 m/s

Marzano check - circle: ① ② ③

04 Apply the physics

Three carts undergo separate bumper tests. Cart A is **1 kg** at **+2 m/s**; B is **3 kg** at **+2 m/s**; C is **1 kg** at **+4 m/s**. Each stops in **0.50 s**. **Find the signed momentum change** and average force in each trial.

Cart	Mass (kg)	Initial v (m/s)	Final v
A	1	2	0
B	3	2	0
C	1	4	0

Marzano check - circle: ① ② ③

What makes motion hard to stop?

antocciphysics

Continue down each column.

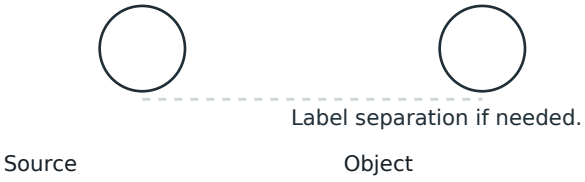
05 Explain your reasoning

Two objects have equal momentum but strike different materials over different times. Can equal momentum alone establish equal damage? Explain **0**.

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

Two equal **4000 kg** masses are **6 m** apart. Use $G=\mathbf{6.67\times10^{-11}\ N\cdot m^2/kg^2}$. **Find force**. By what factor is an answer wrong if r rather than r^2 is used?
Label the masses, field or force relevant to the question.



Marzano check - circle: ① ② ③

07 Exit check

A **3 kg** cart travels right at **4 m/s** and stops in **4 s**. Take right as positive.
1. **Find the signed impulse** and average force.
2. **Draw arrows** for the initial momentum and stopping force.

Cart stopping

Right is positive.



Velocity:

Acceleration / force:

Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

A **0.5 kg** cart at **+8 m/s** and a **2 kg** cart at **+2 m/s** each stop in **0.2 s**. **Compare their momentum changes** and average stopping forces.

Object / trial	Mass	Velocity	Momentum
A	____	____	____
B	____	____	____

Marzano check - circle: ① ② ③